

LIKELIHOOD RATIO TESTS FOR SOME MULTIVARIATE HYPOTHESES WITH RESTRICTED ALTERNATIVES* **

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1. Introduction

Let the columns of the random matrices X ($p \times n_1$), Y ($p \times n_2$) and Z ($p \times n_3$) be independently multinormally distributed with common covariance matrix Σ and expectations $EX = \Lambda$, $EY = 0$ and $EZ = \Lambda_1$. This general MANOVA model will be denoted by Ω . The likelihood ratio (LR) statistics for testing the null hypothesis $H_0: \Lambda = 0$ against the unrestricted alternative $\Omega - H_0$ is Wilks' statistic $W = \prod_{i=1}^p (1 - d_i)$, where the d_i 's are the characteristic roots of $XX'(XX' + YY')^{-1}$ (see e.g. Anderson (1958)). It is well known that the distribution of the d_i 's depends only on the characteristic roots γ_j of $\Sigma^{-1}\Lambda\Lambda'$.

The main purpose of this paper is to derive (in section 3) the LR test of H_0 (which can be redefined as $H_0: \gamma_1 = \gamma_2 = \dots = \gamma_p = 0$) against the restricted alternative $\omega_\rho = \Omega \cap H_\rho$, where $H_\rho: \gamma_j = \rho_j \gamma_1$, ($j = 2, 3, \dots, p$), and $(\rho_2, \rho_3, \dots, \rho_p)$ is a given vector of root ratios. Such hypotheses are of interest when Σ is approximately known up to a proportionality factor, or when the "between-treatments" and "within-treatments" correlation coefficients are approximately known. Other practical applications are connected with situations in which the root ratios can be interpreted as weights given to successive discriminant functions, or to successive orthogonal contrasts in the multivariate treatment means. For a discussion of these applications and interpretations, see Genizi (1969, 1970).

Pillai et al. (1969) have also considered hypotheses about root ratios in this and other multivariate problems. They suggested using the ratios of the corresponding sample characteristic roots as test statistics, and derived their distribution for some simple cases. In particular, they considered in detail the model (which will be denoted by Ω_1) of two independent multinormal random samples X_1 ($p \times n_1$) and X_2 ($p \times n_2$) taken from a $N(0, \Sigma^{(1)})$ and a $N(0, \Sigma^{(2)})$ population respectively, and the problem of testing $H_0: \Sigma^{(2)} = \Sigma^{(1)}$ against $\omega_\delta = H_\delta \cap \Omega_1$, where $H_\delta: \Sigma^{(2)} = \delta \Sigma^{(1)}$. In section 4 we derive the LR test for this problem, and comment on another problem involving a similar estimating

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equation.

The LR tests derived in this paper may be expected to have good power in cases where the corresponding restrictions on the alternatives are known to hold exactly or approximately. For example, the test of section 3 with $p=2$ and $\rho_2=1$ may be useful when γ_1 and γ_2 are known to be approximately equal (this occurs e.g. when the "between-treatments" and "within-treatments", covariance matrices are known to be approximately proportional). This solution in which none of the standard MANOVA tests are efficient, is discussed in detail by Genizi (1969), who also gives tables of approximate .01 and .05 critical points for the test in this case.

The LR tests may also be regarded as a guiding device for choosing between several possible tests of the null hypothesis where the alternatives are known to be in some broadly defined regions. In such a situation one can choose the test whose critical region boundary is "nearest" to the corresponding LR test. Evidence that such choices yield high-power tests is contained in the results of some Monte Carlo studies done by Schatzoff (1966) and Genizi (1969), and of direct power computations by Pillai and Jayachandran (1967).

2. Some notations and lemmas

We shall use the notation $\text{ch}_i(A)$ for the i -th largest characteristic root of the square matrix $A(p \times p)$, (i.e., $\text{ch}_1(A) \geq \text{ch}_2(A) \geq \dots \geq \text{ch}_p(A)$), $\text{tr } A$ for the trace of A , and $\text{etr } A$ for $e^{\text{tr } A}$. For any vector $(\alpha_1, \alpha_2, \dots, \alpha_p)$, D_α will stand for $\text{diag}(\alpha_1, \alpha_2, \dots, \alpha_p)$. If $A(p \times p)$ is non-negative-definite (n.n.d.), then $A^{\frac{1}{2}}(p \times p)$ will denote the symmetric square root of A (see e.g. Bellman (1960), p. 92).

If $B(p \times n)$ is a real matrix and $b_i = \text{ch}_i^{\frac{1}{2}}(BB')$, $i=1, 2, \dots, p$, then there exist two matrices $\Gamma(p \times p)$ and $\Delta(n \times p)$ such that $\Gamma'\Gamma = \Delta'\Delta = I_p$ and $B = \Gamma'D_b\Delta'$ (Deemer et al. (1951)). If $n=p$, put $C = (BB')^{\frac{1}{2}}$ and $G = \Gamma'\Delta'$. Then $B = CG$, and for any vector e of unit length we have $e'Be = e'CGe \leq (e'CC'e)^{\frac{1}{2}}(e'G'G'e)^{\frac{1}{2}} = (e'BB'e)^{\frac{1}{2}}$. If $(e_1, e_2, \dots, e_p)$ is an orthonormal set of characteristic vectors of BB' , we get $\text{tr } B = \sum_{i=1}^p e_i'Be_i \leq \text{tr } (BB')^{\frac{1}{2}}$. Hence, for B and G of order $(p \times p)$,

$$(2.1) \max_{GG'=I} \text{tr } (BG) = \text{tr } (BB')^{\frac{1}{2}}$$

Let $Q(p \times p)$, $X(p \times n)$ and $\Delta(n \times p)$ be matrices such that Q is positive definite (p.d.) and X is of rank p . Then it can be easily seen that

$$\max_{\Delta'\Delta=I} \text{tr } (\Delta'X'QX\Delta)^{\frac{1}{2}} = \sum_{i=1}^p \lambda_i,$$

where $\lambda_i = \text{ch}_i(X'QX)^{\frac{1}{2}} > 0$ ($i=1, 2, \dots, p$). The orthonormal columns of the maximizing matrix Δ_0 are thus restricted to the eigenspace of the non-zero roots of $(X'QX)$, which

implies that $X\Delta_0$ is non-singular. More generally, if $D_\alpha = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_p)$ with $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_p > 0$, then (Horn 1950))

$$(2.2) \max_{\Delta' \Delta = I} \text{tr} [D_\alpha \Delta' X' Q X \Delta D_\alpha]^{\frac{1}{2}} = \sum_{i=1}^p \alpha_i \lambda_i,$$

which again establishes the non-singularity of $X\Delta_0$. Moreover, if $f(e^x)$ is a convex non-decreasing function of x , then (Horn (1950))

$$(2.3) \sum_{i=1}^p f(\mu_i) \leq \sum_{i=1}^p f(\alpha_i \lambda_i),$$

where

$$\mu_i = \text{ch}_i [D_\alpha \Delta' X' Q X \Delta D_\alpha]^{\frac{1}{2}}.$$

We shall use the notations and results of Dwyer (1967) for matrix derivatives. In this notation, the symbol $\langle X \rangle_{ij}$ stands for the "matrix value" of the element x_{ij} of a matrix X as dependent on its position, rather than for its "scalar value". If $y = y(X)$ is a function of X , it will be regarded as a function of its matrix values. Thus, the derivative $\langle \partial y / \partial X \rangle_{ij} = \partial \langle y \rangle / \partial x_{ij}$ will be computed ignoring for the present restrictions imposed on X , such as symmetry or p.d. Those restrictions may be imposed after a maximum is reached by this method of derivation in the class of general matrices.

If A and X are square matrices of the same order, then $\partial \text{tr} AX / \partial X = A'$. If X is p.d., then $\partial \log |X| / \partial X = X^{-1}$. We shall also need the derivatives of $\text{tr} X^{\frac{1}{2}}$ and of $\text{tr} (AXA')^{\frac{1}{2}}$, which can be derived as follows. If X is p.d. and $U = X^{\frac{1}{2}} = U(X)$, we have $X = U(X^2)$ and hence (by the chain rule for matrix differentiation)

$$\begin{aligned} I &= \frac{\partial \text{tr} X^{\frac{1}{2}}}{\partial X^{\frac{1}{2}}} = \frac{\partial \text{tr} U(X)}{\partial X^{\frac{1}{2}}} = \sum_{r,s} \left\langle \frac{\partial \text{tr} U(X)}{\partial X} \right\rangle_{rs} \frac{\partial \langle X \rangle_{rs}}{\partial X^{\frac{1}{2}}} \\ &= \sum_{r,s} \left\langle \frac{\partial \text{tr} U(X)}{\partial X} \right\rangle_{rs} (K_{rs} X^{\frac{1}{2}} + X^{\frac{1}{2}} K_{rs}), \end{aligned}$$

where K_{rs} is a matrix of the same dimension as X , with 1 at the (r,s) position and zeroes elsewhere. It is easily seen that $\partial \text{tr} U(X) / \partial X = \frac{1}{2} X^{-\frac{1}{2}}$ solves this equation, and hence

$$\partial \text{tr} X^{\frac{1}{2}} / \partial X = \frac{1}{2} X^{-\frac{1}{2}}.$$

If A is non singular, then a similar argument gives

$$(2.4) \partial \text{tr} (AXA')^{\frac{1}{2}} / \partial X = \frac{1}{2} A' (AXA')^{-\frac{1}{2}} A.$$

3. LR test of H_0 against ω_p

Consider the general model Ω described in the Introduction, with the added

assumptions that Σ is p.d. and $n_2 \gg p$, which imply that YY' is also p.d. with probability 1. Without loss of generality (Anderson (1958), p. 193) we shall also assume that $n_1 \gg p$. Denote $N = n_1 + n_2 + n_3$, $S_h = XX'$, $S = XX' + YY'$, $\Sigma_h = \Lambda\Lambda'$, and $T = (\Sigma^{-1})^{\frac{1}{2}} = \Sigma^{-\frac{1}{2}}$. Thus, $\gamma_j = \text{ch}_j(\Sigma^{-1}\Sigma_h) = \text{ch}_j(T\Lambda\Lambda'T')$, $j=1,2,\dots,p$.

We start with the following reparametrization of the problem. H_ρ can be written in the form

$$(3.1) \quad H_\rho : \gamma_j = \lambda \alpha_j^2 \quad (j=1,2,\dots,p)$$

where the known constants α_j^2 satisfy $\sum_{j=1}^p \alpha_j^2 = 1$; thus,

$$(3.2) \quad \sum_{j=1}^p \gamma_j = \lambda,$$

and α_j^2 is the known ratio between γ_j and the unknown trace λ . (The null hypothesis H_0 corresponds to $\lambda = 0$.) This in turn, is equivalent to the following statement: there exist matrices $\Gamma(p \times p)$ and $\Delta(n_1 \times p)$ with orthonormal columns such that

$$(3.3) \quad H_\rho : \Lambda = \lambda^{\frac{1}{2}} T^{-1} \Gamma' D_\alpha \Delta'$$

(see Section 2). For given D_α , we can rewrite the likelihood L by putting T^{-2} for Σ and the r.h. side of (3.3) for Λ . Our reparametrization involves a transformation of a set of parameter points (Λ, Σ) into a set of parameter points $(\lambda, \Gamma, \Delta, T)$. Specifically, whenever λ , Γ , Δ and T are such that $\lambda \geq 0$, the columns of Γ and Δ are orthonormal and T is positive definite, one can obtain, using (3.3), matrices Λ and Σ which satisfy (3.1). Moreover, different $(\lambda, \Gamma, \Delta, T)$ -points which correspond to the same (Λ, Σ) -point yield the same value of the likelihood function. Hence, maximizing the likelihood over Σ and Λ is equivalent to maximizing it over λ , Γ , Δ and T , subject to the restrictions stated above.

Lemma 1

The maxima of L over Ω , over $\omega_\rho = H_\rho \cap \Omega$, and over $\omega_0 = H_0 \cap \Omega$, are of the form

$$(3.4) \quad \max L = [(2\pi)^p |\hat{\Sigma}_{\omega}|]^{-N/2} \exp(-\frac{1}{2} Np),$$

where ω denotes the parameter space over which the maximum is taken, and $\hat{\Sigma}_{\omega}$ is the maximum likelihood estimate (MLE) of Σ under ω .

Proof:

The lemma will be proved firstly for the spaces Ω and ω_0 , and secondly for ω_ρ .

(i) The spaces Ω and ω_0

The likelihood function is

$$L = [(2\pi)^p |\Sigma|]^{-\frac{1}{2}N} \text{etr} - \{ \frac{1}{2} \Sigma^{-1} [YY' + (X-\Lambda)(X-\Lambda)' + (Z-\Lambda_1)(Z-\Lambda_1)'] \}.$$

The MLE of Λ_1 under Ω and under ω_o (and also under ω_p) is $\hat{\Lambda}_1 = Z$ and hence

$$(3.5) \log L' = \max_{\Lambda_1} \log L = -\frac{1}{2}N \log [(2\pi)^p |\Sigma|] - \frac{1}{2} \text{tr} \{ \Sigma^{-1} [YY' + (X-\Lambda)(X-\Lambda)'] \}.$$

To maximize L' over Ω or over ω_o , we set

$$\frac{\partial \log L'}{\partial \Sigma^{-1}} = \frac{1}{2}N\Sigma - \frac{1}{2}[YY' + (X-\Lambda)(X-\Lambda)'] = O(p \times p).$$

Hence, the MLE of $N\Sigma$ is $N\hat{\Sigma} = YY' + (X-\Lambda)(X-\Lambda)'$, and therefore

$$\text{etr} \{ -\frac{1}{2}\hat{\Sigma}^{-1}[YY' + (X-\Lambda)(X-\Lambda)'] \} = \text{etr} \{ -\frac{1}{2}N I_p \} = \exp(-\frac{1}{2}Np),$$

which proves (3.4) for these cases. The proof of (3.4) for ω_p will be given later.

For simplicity, we shall denote the MLE's under ω_o and ω_p by the suffices o and p respectively, e.g. $\hat{\Sigma}_o$, and $\hat{\Sigma}_p$, and sometimes the suffix will be deleted altogether, when its identity is clear from the context. The expressions for $\hat{\Sigma}_\Omega$ and $\hat{\Sigma}_o$ can be easily obtained from (3.5). Under ω_o , the MLE of Λ is $\hat{\Lambda}_o = 0$, and hence $N\hat{\Sigma}_o = YY' + XX'$. Under Ω , the MLE of Λ is $\hat{\Lambda}_\Omega = X$, and hence $N\hat{\Sigma}_\Omega = YY'$.

(ii) The space ω_p

Under ω_p , by (3.3), Λ is a function of Σ , and the direct differentiation method does not work. We shall derive the maximum of L' under H_p in four steps, by maximizing successively over λ , Γ , Σ and Δ .

(a) Maximization over λ , given Γ , Σ and Δ

In view of (3.5), L' will be maximized when $\text{tr} \{ \Sigma^{-1} [YY' + (X-\Lambda)(X-\Lambda)'] \}$ is minimized, i.e. when $\text{tr} \{ \Sigma^{-1} (X-\Lambda)(X-\Lambda)' \} = Q$ (say), is minimized. Using (3.2) and (3.3), we obtain

$$\begin{aligned} Q &= \text{tr} (\Sigma^{-1} XX') - 2\lambda^{1/2} \text{tr} (\Gamma \Gamma' D_\alpha \Delta' X') + \lambda \\ &= \text{tr} (\Sigma^{-1} XX') - \text{tr}^2 (\Gamma \Gamma' D_\alpha \Delta' X') + [\text{tr} (\Gamma \Gamma' D_\alpha \Delta' X') - \lambda^{\frac{1}{2}}]^2. \end{aligned}$$

This is minimized over λ when

$$(3.6) + \lambda^{\frac{1}{2}} = \text{tr} (\Gamma \Gamma' D_\alpha \Delta' X'),$$

$$\text{giving } \log L_1 = \max_{\lambda|\Gamma, \Sigma, \Delta, \omega_\rho} \log L' = -\frac{1}{2}N \log [(2\pi)^p |\Sigma|] - \text{tr } \Sigma^{-1}S + \text{tr}^2(T\Gamma' A),$$

where $A = D_\alpha \Delta' X'$.

(b) *Maximization over Γ , given Σ and Δ*

In view of (2.1),

$$\begin{aligned} \max \{ \text{tr } (\Gamma' A T) | \Gamma(p \times p), \Gamma \Gamma' = I_p \} &= \text{tr } (A T T' A')^{\frac{1}{2}} \\ &= \text{tr } (A \Sigma^{-1} A')^{\frac{1}{2}}. \text{ Hence,} \end{aligned}$$

$$(3.7) \quad \log L_2 = \max_{\Gamma|\Sigma, \Delta, \omega_\rho} \log L_1 = -\frac{1}{2}N \log [(2\pi)^p |\Sigma|] - \frac{1}{2} \text{tr } (\Sigma^{-1}S) + \frac{1}{2} \text{tr}^2 (A \Sigma^{-1} A')^{\frac{1}{2}}.$$

(c) *Maximization over Σ , given Δ*

Consider first the case $\alpha_i^2 > 0$ ($i=1, \dots, p$). (The case where $\alpha_i = 0$ for some i requires only a minor modification, as explained in Remark (e) at the end of this section.) From the results given in Section 2 it follows that we may assume that A is non-singular (with probability 1), because this does not restrict the maximization of L_2 . To maximize over Σ , we set

$$\frac{\partial \log L_2}{\partial \Sigma^{-1}} = \frac{1}{2} \hat{N} \Sigma - \frac{1}{2} S + \frac{1}{2} A' (\hat{A} \hat{\Sigma}^{-1} A')^{-\frac{1}{2}} A \text{tr } (\hat{A} \hat{\Sigma}^{-1} A')^{\frac{1}{2}} = 0.$$

Multiplication of this by $2(\hat{N} \hat{\Sigma})^{-1}$ from the left yields

$$(3.8) \quad (\hat{N} \hat{\Sigma})^{-1} S - I = (\hat{N} \hat{\Sigma})^{-1} A' [A (\hat{N} \hat{\Sigma})^{-1} A']^{-\frac{1}{2}} A \text{tr } [A (\hat{N} \hat{\Sigma})^{-1} A']^{\frac{1}{2}},$$

and hence

$$(3.9) \quad \text{ch}_i[(\hat{N} \hat{\Sigma})^{-1} S - I] = \text{ch}_i[A (\hat{N} \hat{\Sigma})^{-1} A']^{\frac{1}{2}} \text{tr } [A (\hat{N} \hat{\Sigma})^{-1} A']^{\frac{1}{2}}.$$

Summation over i gives

$$(3.10) \quad \text{tr}[(\hat{N} \hat{\Sigma})^{-1} S - I] = \text{tr}^2[A (\hat{N} \hat{\Sigma})^{-1} A']^{\frac{1}{2}},$$

and substituting (3.10) into (3.9), we get

$$(3.11) \quad \text{ch}_i[A (\hat{N} \hat{\Sigma})^{-1} A']^{\frac{1}{2}} = \text{ch}_i[(\hat{N} \hat{\Sigma})^{-1} S - I] / \text{tr}^{\frac{1}{2}}[(\hat{N} \hat{\Sigma})^{-1} S - I].$$

To obtain $\hat{\Sigma}$ we multiply (3.8) by $S^{-1}A'$ from the right and by A from the left, which yields, after some manipulations

$$(AS^{-1}A')^{-1} = [A(N\hat{\Sigma})^{-1}A']^{-1} + [A(N\hat{\Sigma})^{-1}A']^{-\frac{1}{2}} \text{tr}[A(N\hat{\Sigma})^{-1}A']^{\frac{1}{2}}.$$

Hence using (3.10) and (3.11), we have

$$(3.12) \quad \frac{1}{g_i^2} = \left[\frac{1}{(\hat{\theta}_i^2 - 1)^2} \sum_{j=1}^p (\hat{\theta}_j^2 - 1) + \frac{1}{\hat{\theta}_i^2 - 1} \sum_{j=1}^p (\hat{\theta}_j^2 - 1) \right]$$

$$= \frac{\hat{\theta}_i^2}{(\hat{\theta}_i^2 - 1)^2} \sum_{j=1}^p (\hat{\theta}_j^2 - 1),$$

where

$$g_i^2 = \text{ch}_i(AS^{-1}A') \text{ and } \hat{\theta}_i^2 = \text{ch}_i[(N\hat{\Sigma})^{-1}S].$$

(3.12) is an implicit equation connecting the unknowns $\hat{\theta}_i$ (which determine $|\Sigma|$) with the unknowns g_i (which determine the maximizing Δ).

Proof of Lemma 1 for ω_ρ

Substituting (3.10) into (3.7), we have

$$(3.13) \quad \log L_3 = \max_{\Sigma|\Delta, \omega_\rho} \log L_2 = -\frac{1}{2}N \log [(2\pi)^p |\Sigma_\rho|] - \frac{1}{2}Np.$$

(d) *Maximization over Δ*

By (3.13), maximizing L_3 is equivalent to minimizing $|\Sigma|$, or to maximizing $\prod_{i=1}^p \hat{\theta}_i^2 = |(N\hat{\Sigma})^{-1}S|$. First, we maximize this for a given value of

$$(3.14) \quad \hat{\epsilon}^2 = \sum_{i=1}^p (\hat{\theta}_i^2 - 1).$$

From (3.12), we have

$$(3.15) \quad \hat{\theta}_i = h(g_i) = \frac{1}{2}g_i\hat{\epsilon} + \left(\frac{1}{4}g_i^2\hat{\epsilon}^2 + 1\right)^{\frac{1}{2}},$$

($i=1,2,\dots,p$). It can be verified by a somewhat lengthy calculation of the second derivatives, that $f(e^{\hat{g}}) = \log h(e^{\hat{g}})$ is non-decreasing and convex in g . Therefore, according

to the known results given in Section 2, $\log \prod_{i=1}^p \hat{\theta}_i = \sum_{i=1}^p \log h(g_i) = \sum_{i=1}^p f(g_i)$ is maximized (for given $\hat{\epsilon}$) when $g_i^2 = \alpha_i^2 d_i$. But, using (3.14), summation of (3.15) over i gives

$$\sum_{i=1}^p g_i^2/2 + \sum_{i=1}^p \{g_i^4/4 + g_i^2/\hat{\epsilon}^2\}^{\frac{1}{2}} = 1,$$

showing that $\hat{\epsilon}$ is completely determined by the g_i 's. Thus, starting from any set of permissible values for the g_i 's, and using (3.14) and (3.15) for an iterative solution for the $\hat{\theta}_i$'s and $\hat{\epsilon}$, we can apply the results of section 2 and arrive at the new set $g_i^2 = \alpha_i^2 d_i$, which (together with the corresponding new values of $\hat{\epsilon}^2$ and of the $\hat{\theta}_i$'s) maximizes $\Pi \hat{\theta}_i$. This argument together with Lemma 1, completes the proof of

Theorem 1:

Under the model Ω , the LR statistic for testing the null hypothesis $H_0: \gamma_1 = \gamma_2 = \dots = \gamma_p = 0$ against the restricted alternative $H_p: \gamma_i = \lambda \alpha_i^2$ ($i=1,2,\dots,p$), for given constants α_i^2 satisfying $\alpha_1^2 \geq \alpha_2^2 \geq \dots \geq \alpha_p^2 > 0$ and $\sum_{i=1}^p \alpha_i^2 = 1$, is $\Psi^{2/N} = |N\hat{\Sigma}_p|/|N\hat{\Sigma}_0| = |N\hat{\Sigma}_p S^{-1}| = \prod_{i=1}^p \hat{\theta}_i^{-2}$, where $\hat{\theta}_1, \dots, \hat{\theta}_p$ is obtained from

$$(3.16) \quad \hat{\theta}_i = (\frac{1}{4} \alpha_i^2 d_i \hat{\epsilon}^2)^{\frac{1}{2}} + (\frac{1}{4} \alpha_i^2 d_i \hat{\epsilon}^2 + 1)^{\frac{1}{2}}, \quad (i=1,2,\dots,p)$$

and $\hat{\epsilon}^2$ has to satisfy the equation

$$(3.17) \quad \sum_{i=1}^p \alpha_i^2 d_i / 2 + \sum_{i=1}^p \{(\alpha_i^2 d_i / 2)^2 + \alpha_i^2 d_i / \hat{\epsilon}^2\}^{\frac{1}{2}} = 1,$$

which can be solved iteratively.

Note that the left-hand side of (3.17) is monotonically decreasing in $\hat{\epsilon}^2$ from ∞ (for $\hat{\epsilon}^2 = 0$) to $\sum_{i=1}^p \alpha_i^2 d_i \leq d_1 \leq 1$ (for $\hat{\epsilon}^2 = \infty$), and therefore there is one and only one $\hat{\epsilon}^2$ satisfying it. An initial value of $3\alpha_1^2 d_1$ for $\hat{\epsilon}^2$ and increments of $\alpha_1^2 d_1$ will in general result in quite rapid convergence.

The following additional remarks are discussed in detail by Genizi (1969).

(a) In the case $p = 2$, the equations (3.16) and (3.17) can be solved explicitly, yielding the expression

$$(3.18) \quad \prod_{i=1}^2 \hat{\theta}_i^{-1} = \{(1 - \alpha_1^2 d_1)(1 - \alpha_2^2 d_2)\}^{\frac{1}{2}} - \{\alpha_1^2 d_1 \alpha_2^2 d_2\}^{\frac{1}{2}}.$$

for the LR statistic.

(b) If we replace the constants α_i^2 in (3.16) by their estimates under Ω (i.e., $\alpha_i^2 = c_i / \sum_{j=1}^p c_j$, where $c_i = \text{ch}_i(|XX'(YY')^{-1}|)$, we obtain $\Psi^{2/N} = \prod_{i=1}^p (1+c_i)^{-1} = \prod_{i=1}^p (1-d_i)$, which is the LR statistic for H_0 under Ω .

(c) Using (3.6), (3.7) and (3.10), it can be easily shown that the MLE of $\lambda = \sum_{i=1}^p \gamma_i$ under ω_ρ is $\hat{\lambda}_\rho = \text{tr}[(N\hat{\Sigma})^{-1}S-1] = \sum_{j=1}^p (\hat{\theta}_j^2 - 1) = \hat{\epsilon}^2$. This also gives the MLE $\hat{\gamma}_{i\rho} = \alpha_i^2 \hat{\lambda}_\rho$

of $\gamma_i = \lambda \alpha_i^2$.

(d) The LR statistic for testing H_ρ under Ω is $\Psi_1^{2/N} = |\hat{N}\hat{\Sigma}_\Omega|/|\hat{N}\hat{\Sigma}_\rho| = (N\hat{\Sigma}_\rho)^{-1}YY' = \prod_{i=1}^p \hat{\theta}_i^2(1-d_i)$. In the case $p = 2$, this yields a confidence interval for ρ . $= \alpha_2/\alpha_1$, in the form $|\rho^{\frac{1}{2}} - (c_1 c_2 W_\alpha^2)^{\frac{1}{2}}| \leq [c_1 c_2 W_\alpha^2 - (1+c_1-W_\alpha^2)(1+c_2-W_\alpha^2)]^{\frac{1}{2}}$, where W_α^2 is the upper $(1-\alpha)$ quantile of Ψ_1 . Approximate values of W_α can be computed from the asymptotic χ^2 distribution of $-2\log \Psi_1$ with $p-1$ d.f., or from a closer approximation to its distribution using the moments of $\tau = (\alpha_1 \alpha_2)^{-1}$ which are given by Genizi [(1969), p. 59]. Incidentally, the statistic $-2\log \Psi$ itself is asymptotically distributed as χ^2 with $n_1 p - (p-1)$ d.f., and therefore yields a higher asymptotic power under H_ρ than the unrestricted LR test statistic $-2\log(|N\hat{\Sigma}_\Omega|/|N\hat{\Sigma}_0|)^{N/2}$, which has $p-1$ d.f. more and yet the same non-centrality parameter (Aitchison 1962).

(e) The case where some α_i vanish can be handled as follows. If $\alpha_{h+1}^2 = \alpha_{h+2}^2 = \dots = \alpha_p^2 = 0$, put $H^{(h)}: \gamma_{h+1} = \dots = \gamma_p = 0$. The LR statistic for testing H_0 under H_ρ is the product of the LR statistic for H_0 under $H^{(h)}$ (which has been derived by Anderson (1945)), and the LR statistic for $H^{(h)}$ under H_ρ . Using this device, it can be easily shown that $\Psi^{2/N} = \prod_{i=1}^h \hat{\theta}_i^{-2}$, where now the $\hat{\theta}_i$ are obtained from (3.16) and (3.17) with p replaced by h .

4. Some other L.R. tests against restricted alternatives

(i) A LR test for Ω_1 .

Consider the model Ω_1 described in the Introduction. The log-likelihood L under $H_\delta: \Sigma^{(1)} = \delta^{-1} \Sigma$ is given by

$$\log L = -\frac{1}{2}Np\log(2\pi) + \frac{1}{2}n_1p\log\delta - \frac{1}{2}N\log|\Sigma| - \frac{1}{2}\text{tr}[\Sigma^{-1}(\delta X_1 X_1' + X_2 X_2')],$$

where $N = n_1 + n_2$. Differentiation with respect to δ and Σ , along the lines of section 3, yields the LR statistic

$$(4.1) \quad \Psi_2^{2/N} = |\hat{N}\hat{\Sigma}_\delta|/|\hat{N}\hat{\Sigma}_0| = \frac{|\delta X_1 X_1' + X_2 X_2'|}{|X_1 X_1' + X_2 X_2'|} = \prod_{i=1}^p \frac{c_i + \delta}{c_i + 1},$$

where $c_i = \text{ch}_i[X_1' X_1]^{-1} (X_2' X_2)$, and $\hat{\delta}$ is that solution of

$$(4.2) \quad n_1 p \hat{\delta}^{-1} = N \sum_{i=1}^p (c_i + \hat{\delta})^{-1}$$

which minimizes $|(X_1' X_1)^{-1} (N \hat{\Sigma}_{\hat{\delta}})| = \prod_{i=1}^p (c_i + \hat{\delta})$.

We have computed iterative solutions of (4.2) for some simple cases. These computations indicate that an iterative procedure starting with the initial value $\hat{\delta} = 1$ on the r.h. side of (4.2) — will in many cases (and perhaps always) lead to the minimizing solution.

(ii) *A LR test of a hypothesis specifying the trace in Ω .*

It is interesting to note that the equation (4.2) for the MLE of $\hat{\delta}$ is similar to that obtained in the following quite different situation. In the general linear model Ω of section 1, consider the problem of testing the hypothesis

$$H: \sum_{i=1}^p \gamma_i = \lambda_0,$$

with given λ_0 . (Generalization to the related problems of testing $\sum_{i=1}^p \gamma_i \geq \lambda_0$ or $\sum_{i=1}^p \gamma_i \leq \lambda_0$ is straightforward.)

To obtain the LR test, we have to maximize the expression

$$\log \phi = \log L' - \frac{1}{2} \mu (\text{tr} \Sigma^{-1} \Lambda \Lambda' - \lambda_0),$$

where μ is a lagrangian multiplier and $\log L'$ is given by (3.5). This can be done by setting

$\partial \log \phi / \partial \Sigma^{-1} = 0$ and then maximizing over Λ which gives

$$(4.3) \quad \hat{\Lambda}_{\lambda} = \hat{\delta}^{-1} X \text{ and } N \hat{\Sigma}_{\lambda} = Y Y' + X X' (\hat{\delta} - 1) / \hat{\delta}$$

where $\hat{\delta} = \hat{\delta}_{\lambda} = \hat{\mu}_{\lambda} + 1$, the suffix λ indicating that the MLE is under ω_{λ} . Hence, $|N \hat{\Sigma}_{\lambda} S^{-1}| = \prod_{i=1}^p (1 - d_i / \hat{\delta})$, and

$$(4.4) \quad \max_{\omega_{\lambda}} \log L' = -\frac{1}{2} N p \log (2\pi) - \frac{1}{2} N \log |\hat{\Sigma}_{\lambda}| + \frac{1}{2} (\hat{\delta} - 1) \lambda_0 - \frac{1}{2} N p.$$

To obtain $\hat{\delta}$, it is convenient to derive first the estimators of γ_i from (4.3), i.e.,

$$\begin{aligned}\hat{\gamma}_i &= \text{ch}_i(\hat{\Sigma}^{-1} \Lambda \Lambda') = N \hat{\delta}^{-2} \text{ch}_i \{ [YY' + XX'(\hat{\delta} - 1)/\hat{\delta}]^{-1} XX' \} \\ &= N \hat{\delta}^{-1} d_i / (\hat{\delta} - d_i)\end{aligned}$$

Now, putting $\sum_{i=1}^p \gamma_i = \lambda_0$, we see that $\hat{\delta}$ is that root of the equation

$$(4.5) \quad \lambda_0 \hat{\delta} = N \sum_{i=1}^p d_i / (\hat{\delta} - d_i)$$

which maximizes (4.4). However, in this case it has been shown (Genizi 1969)) that only the (unique) solution satisfying $\hat{\delta} > d_1$ gives proper values of (4.4), and this solution can be easily obtained by graphical or iterative methods.

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